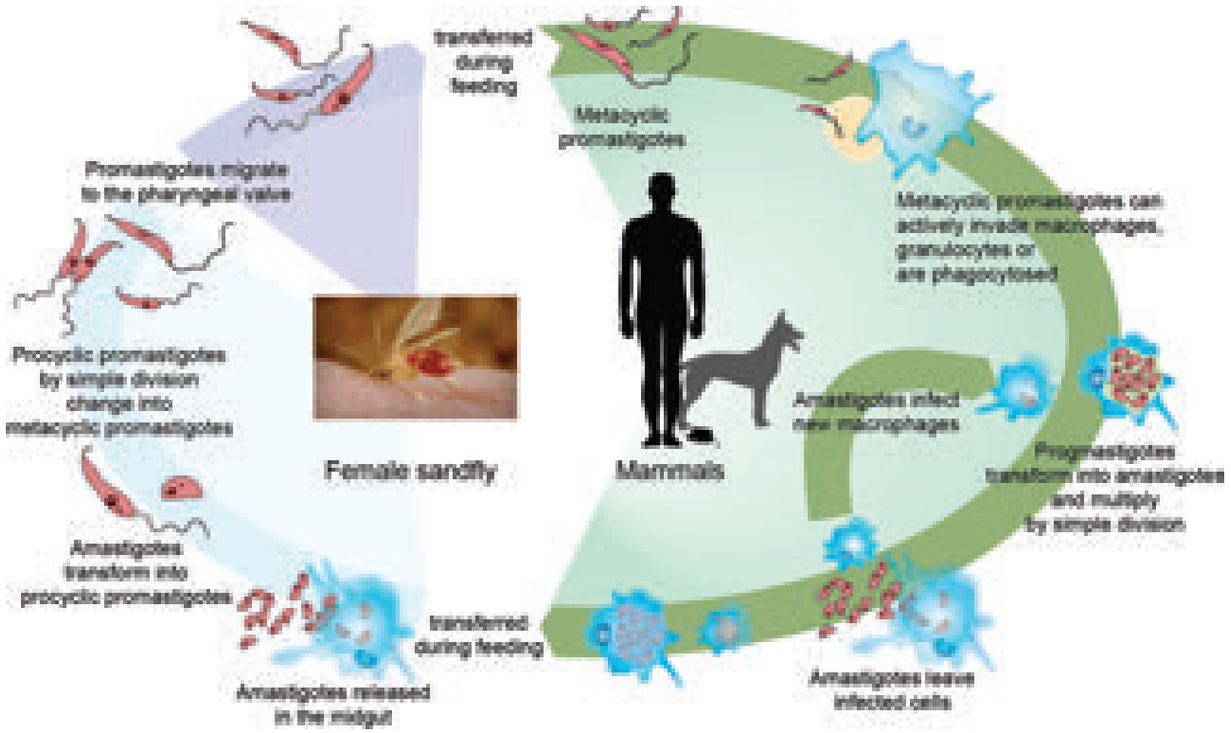


Figure 8.9: Life cycle of *Leishmania donovani*

These promastigotes multiply by binary fission. After a period of 6 to 9 days, these forms migrate from the midgut to the pharynx and buccal cavity of sandfly. Bite of the infected sandfly transmits infection to susceptible persons and the life – cycle is repeated.

8.4.5 Pathogenesis

Leishmania donovani causes visceral Leishmaniasis. The disease is also known as Dum – Dum fever, Asian fever, Assam fever, or infantile splenomegaly. Leishmaniasis is a disease of the reticuloendothelial system. Proliferation and destruction of reticuloendothelial cells of the internal organs are responsible for the pathological changes in visceral leishmaniasis.

Spleen, liver and lymphnodes are enlarged in this condition. Bone marrow is dark red in colour and shows extensive proliferation of reticuloendothelial cells. Kidney shows cloudy swelling and is invaded by macrophages parasitized by amastigotes.

8.4.6 Clinical Features

Incubation period: It is usually 3–6 months but can be months or years.

Visceral Leishmaniasis is a serious and fatal systemic disease. In India, the disease is called Kala – azar meaning "black disease".

The disease is characterized by the presence of fever, hepatosplenomegaly (Figure 8.10) (the simultaneous enlargement of both liver and the spleen), hypergammaglobulinemia (a condition in which increased levels of a certain immunoglobulin in blood serum), Leucopenia, Thrombocytopenia (deficiency of platelets in the blood), Cachexia (a condition that causes extreme weight loss)

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